Planetary Gear System N=3

Rudhdam Shende

August 2026

1 Calculations

1.1 Tooth Count Compatibility

For a simple planetary gear system:

$$N_r = N_s + 2N_p \tag{1}$$

$$N_r = 20 + 2(22) = 64 \tag{2}$$

Therefore,

$$N_s = 20, \qquad N_p = 22, \qquad N_r = 64 \tag{3}$$

For three planets, the assembly condition is:

$$\frac{N_s + N_r}{N} = \frac{20 + 64}{3} = 28 \tag{4}$$

Hence, the three-planet arrangement is compatible.

1.2 Sun Gear (N_s = 20 Teeth)

$m = 2$ mm

Pitch Diameter

$$d_s = mN_s \tag{5}$$

$$d_s = 2(20) = 40.000 \text{ mm} \tag{6}$$

$$r_s = 20.000 \text{ mm} \tag{7}$$

Addendum

$$h_a = m \tag{8}$$

$$h_a = 2.000 \text{ mm} \tag{9}$$

Dedendum

$$h_f = 1.25m \tag{10}$$

$$h_f = 2.500 \text{ mm} \tag{11}$$

Outside Diameter

$$d_{a,s} = m(N_s + 2) \tag{12}$$

$$d_{a,s} = 44.000 \text{ mm} \tag{13}$$

Root Diameter

$$d_{f,s} = m(N_s - 2.5) \tag{14}$$

$$d_{f,s} = 35.000 \text{ mm} \tag{15}$$

Base Diameter

$$d_{b,s} = d_s \cos(20^\circ) \tag{16}$$

$$d_{b,s} = 37.588 \text{ mm} \tag{17}$$

$$r_{b,s} = 18.794 \text{ mm} \tag{18}$$

TOOTH THICKNESS AT PITCH CIRCLE

$$t = s = \frac{\pi m}{2} \tag{19}$$

$$t = 3.1416 \text{ mm} \tag{20}$$

CIRCULAR PITCH

$$p = \pi m \tag{21}$$

$$p = 6.283 \text{ mm} \tag{22}$$

ANGULAR PITCH

$$\theta = \frac{360^\circ}{N_s} \tag{23}$$

$$\theta = 18^\circ \tag{24}$$

1.3 Planet Gear ($N_p = 22$ Teeth)

$$N_p = 22 \tag{25}$$

PITCH DIAMETER

$$d_p = mN_p \tag{26}$$

$$d_p = 44.000 \text{ mm} \tag{27}$$

$$r_p = 22.000 \text{ mm} \tag{28}$$

ADDENDUM / DEDENDUM

$$h_a = 2.000 \text{ mm} \quad \text{(Addendum)} \tag{29}$$

$$h_f = 2.500 \text{ mm} \quad \text{(Dedendum)} \tag{30}$$

OUTSIDE DIAMETER

$$d_{a,p} = m(N_p + 2) \tag{31}$$

$$d_{a,p} = 48.000 \text{ mm} \tag{32}$$

$$r_{a,p} = 24.000 \text{ mm} \tag{33}$$

ROOT DIAMETER

$$d_{f,p} = m(N_p - 2.5) \tag{34}$$

$$d_{f,p} = 39.000 \text{ mm} \tag{35}$$

$$r_{f,p} = 19.500 \text{ mm} \tag{36}$$

BASE DIAMETER

$$d_{b,p} = d_p \cos(20^\circ) \tag{37}$$

$$d_{b,p} = 41.346 \text{ mm} \tag{38}$$

$$r_{b,p} = 20.673 \text{ mm} \tag{39}$$

CIRCULAR PITCH

$$p = \pi m \tag{40}$$

$$p = 6.283 \text{ mm} \tag{41}$$

TOOTH THICKNESS

$$t = \frac{\pi m}{2} \tag{42}$$

$$t = 3.1416 \text{ mm} \tag{43}$$

ANGULAR PITCH

$$\phi = \frac{360^\circ}{N_p} \tag{44}$$

$$\phi = 16.3636^\circ \tag{45}$$

1.4 Ring Gear ($N_r = 64$ Teeth)

PITCH DIAMETER

$$d_r = mN_r \tag{46}$$

$$d_r = 2(64) \tag{47}$$

$$d_r = 128.000 \text{ mm} \tag{48}$$

$$r_r = 64.000 \text{ mm} \tag{49}$$

BASE DIAMETER

$$d_{b,r} = d_r \cos(20^\circ) \tag{50}$$

$$d_{b,r} = 120.281 \text{ mm} \tag{51}$$

$$r_{b,r} = 60.140 \text{ mm} \tag{52}$$

For an internal gear, the tooth geometry is inverted relative to an external gear.

INSIDE TIP / TOOTH-TIP DIAMETER

$$d_{a,r} = d_r - 2m \tag{53}$$

$$d_{a,r} = 128 - 2(2) \tag{54}$$

$$d_{a,r} = 124.000 \text{ mm} \tag{55}$$

OUTSIDE / ROOT DIAMETER

$$d_{f,r} = d_r + 2.5m \tag{56}$$

$$d_{f,r} = 128 + 2.5(2) \tag{57}$$

$$d_{f,r} = 133.000 \text{ mm} \tag{58}$$

1.5 Planet Position

Centre Distance:

$$C_{s,p} = \frac{d_s + d_p}{2} \qquad \text{[Sun–Planet]} \tag{59}$$
$$= \frac{40 + 44}{2} \tag{60}$$
$$C_{s,p} = 42.000 \text{ mm} \tag{61}$$

$$C_{p,r} = \frac{d_r - d_p}{2} \qquad \text{[Planet–Ring]} \tag{62}$$
$$= \frac{128 - 44}{2} \tag{63}$$
$$C_{p,r} = 42.000 \text{ mm} \tag{64}$$

Carrier Centre Radius:

$$R_c = 42.000 \text{ mm} \tag{65}$$
$$D_c = 84.000 \text{ mm} \tag{66}$$

1.6 Gear Ratio: Sun–Ring

With the carrier fixed:

$$i = \frac{N_r}{N_s} \tag{67}$$

$$i = \frac{64}{20} \tag{68}$$

$$\mathbf{i = 3.2{:}1} \tag{69}$$

Using the Willis equation:

$$\frac{\omega_s - \omega_c}{\omega_r - \omega_c} = -\frac{N_r}{N_s} \tag{70}$$

For a fixed carrier, $\omega_c = 0$:

$$\frac{\omega_s}{\omega_r} = -\frac{N_r}{N_s} \tag{71}$$

Therefore,
Given:

$$\omega_s = 10 \text{ RPM} \tag{72}$$

Since

$$\omega_r = -\omega_s \frac{N_s}{N_r} \tag{73}$$

$$\omega_r = -10 \frac{20}{64} \tag{74}$$

$$\boxed{\omega_r = -3.125 \text{ RPM}} \tag{75}$$